Using Moreau-Yosida regularization in DFT

Outline

1. DFT & iKS (motivation)
2. MY iKS

1A. Density-Functional theory (DFT)

$E(v)$ interacting gs energy $\quad v \in X^*$

$\hat{H} = \hat{T} + \hat{W} + \hat{V}$ = "kin." + "two-particle" + "ext. pot"

$$F(\rho) = \sup_{v \in X^*} \left(E(v) - \langle v, \rho \rangle \right), \quad \rho \in X$$

$E_{KS}(v_{KS})$ non-interacting gs energy

$\hat{H}_{KS} = \hat{T} + \hat{V}_{KS}$ = "kin." + "effective ext. pot."

$$F_{KS}(\rho) = \sup_{v \in X^*} \left(E_{KS}(v) + \langle v, \rho \rangle \right)$$

X space of densities, Banach space

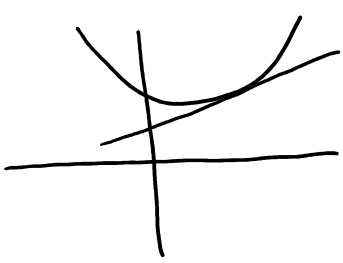
X^* space of potentials, dual space of X

$J: X \to X^*$ duality mapping

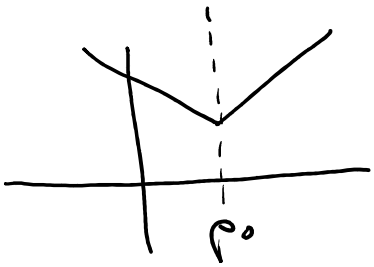
$$J(\rho) = \{ v \in X^* \mid \|v\|^2 = \|\rho\|^2 = \langle v, \rho \rangle \}$$

The subdiff. of $f: X \to \mathbb{R}$

$$\partial f(\rho) = \{ v \in X^* \mid f(\rho') \geq \underbrace{f(\rho) + \langle v, \rho' - \rho \rangle}_{\text{tangent of } f \text{ at } \rho} \ \forall \rho' \in X \}$$



$\partial f \neq \emptyset$



$\partial f(\rho_0)$ not a singleton

General fact: $\partial\left(\frac{1}{2}\|\rho\|^2\right) = J(\rho)$

1B. inverse Kohn-Sham

Let ρ_{ss} be ss particle density

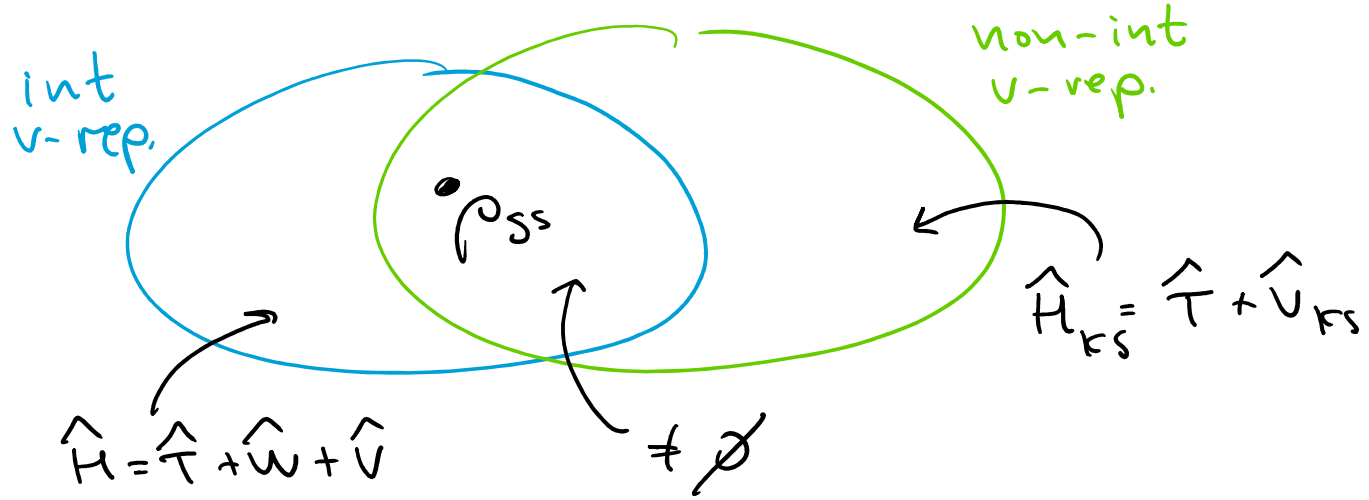
$$\rho_{ss} = \operatorname{argmin}\left(F(\rho) + \langle v, \rho \rangle \right) \qquad v = v_{ext}$$

$$\rho_{ss} = \operatorname{argmin}\left(F_{ks}(\rho) + \langle v_{ks}, \rho \rangle \right) \qquad v_{ks} = \text{eff. pot.}$$

$-v \in \partial F(\rho_{ss})$ ρ_{ss} is ss of $\hat{H} = \hat{T} + \hat{W} + \hat{V}$

$$\boxed{-v_{ks} \in \partial F_{ks}(\rho_{ss}) \neq \emptyset}$$

Assumption of non-interacting v-representability



$$\boxed{v_{ks} = v + v_H + v_{xc}}$$

1C. ZMP method

ρ_{ss} given, need to find v_{xc}

$$\min_{\rho} \left(F_{KS}(\rho) + \langle v, \rho \rangle + D(\rho,\rho) + \underbrace{\lambda D(\rho-\rho_{ss}, \rho-\rho_{ss})}_{\text{penalty term}} \right)$$

$$\rho^\lambda = \sum_\gamma |\phi_\gamma^\lambda|^2 , \quad \{\phi_\gamma^\lambda\} \text{ orbitals}$$

$$\left(-\tfrac{1}{2}\Delta + v + v_H^\lambda + v^\lambda \right) \phi_\gamma^\lambda = \varepsilon_\gamma^\lambda \phi_\gamma^\lambda$$

$$v^\lambda(x) = \lambda \int \frac{\rho(x') - \rho_{ss}(x')}{|x-x'|} dx' \xrightarrow{\text{num.}} v_{xc}(x) \quad \text{as } \lambda \to \infty$$

2A. Moreau-Yosida in DFT

$f : X \to \mathbb{R} \cup \{+\infty\}$ convex and lower semicontinuous: $\{f(\rho) \le \alpha\}$ is closed

$$f_\varepsilon(\rho_0) = \inf_{\rho \in X} \left(f(\rho) + \frac{1}{2\varepsilon} \|\rho - \rho_0\|_X^2 \right)$$

$$(\rho_0)_\varepsilon = \operatorname{argmin}_{\rho \in X} \left(f(\rho) + \frac{1}{2\varepsilon} \|\rho - \rho_0\|_X^2 \right)$$ proximal point of ρ_0

with $f(\rho) = F(\rho) = \sup_{v \in X^*} \left(E(v) - \langle v, \rho \rangle \right) \Rightarrow$

$$E_\varepsilon(v) = \inf_{\rho \in X} \left(F_\varepsilon(\rho) + \langle v, \rho \rangle \right)$$

$$= E(v) - \frac{\varepsilon}{2} \|v\|^2$$ "lossless" regularization

2B. Moreau–Yosida inverse Kohn-Sham

Take $f(\rho) = F_{KS}(\rho) = \sup_{v \in X^*} \left(E_{KS}(v) - \langle v, \rho \rangle \right)$

Let ρ_{ss} be given, X reflexive, strictly convex

$$(\rho_{ss})_\varepsilon = \underset{\varepsilon F_{KS}}{\text{prox}}(\rho_{ss}) = \inf_{\rho \in X} \left(F_{KS}(\rho) + \frac{1}{2\varepsilon} \|\rho - \rho_{ss}\|_X^2 \right)$$

$\rho_{ss} \in \text{dom}(F_{KS})$: $(\rho_{ss})_\varepsilon \xrightarrow[\varepsilon \to 0]{} \rho_{ss}$

$$\begin{cases} \partial F_{KS}((\rho_{ss})_\varepsilon) + \frac{1}{\varepsilon} J((\rho_{ss})_\varepsilon - \rho_{ss}) \ni 0 & \text{stationary condition} \\ \partial F_{KS}(\rho_{ss}) + v_{KS} \ni 0 & \text{assumption of non-int v-rep.} \end{cases}$$

$\Rightarrow$ [technical details] $\Rightarrow$

$$v_{KS}(x) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} J\left((\rho_{ss})_\varepsilon - \rho_{gs} \right)$$

cf. ZMP $\lambda \to \infty$

Ex| $X = L^2$, $X^* = L^2$, $J(\rho) = \rho$

$\Rightarrow$ van Leeuwen & Baerends

Ex| $X = H^{-1}$, $X^* = H^1$ $J(\rho) = \gamma$ with γ Yukawa $* \rho$

$\Rightarrow$

$$v_{xc}(x) = \lim_{\varepsilon \to 0} \frac{1}{4\pi\varepsilon} \int \frac{e^{-\gamma|x-x'|}}{|x-x'|} \left((\rho_{ss})_\varepsilon(x') - \rho_{ss}(x') \right) dx'$$

close to ZMP